quantum effects drive the strength of the electromagnetic force to get larger when examined on shorter distance scales.

What about the other forces of the standard model? How do their intrinsic strengths vary with distance? In 1973, Gross and Frank Wilczek at Princeton, and, independently, David Politzer at Harvard, studied this question and found a surprising answer: The quantum cloud of particle eruptions and annihilations amplifies the strengths of the strong and weak forces. This implies that as we examine them on shorter distances, we penetrate more of this seething cloud and hence are subject to less of its amplification. And so, the strengths of these forces get weaker when they are probed on shorter distances.

Georgi, Quinn, and Weinberg took this realization and ran with it to a remarkable end. They showed that when these effects of the quantum frenzy are carefully accounted for, the net result is that the strengths of all three nongravitational forces are driven together. Whereas the strengths of these forces are very different on scales accessible to current technology, Georgi, Quinn, and Weinberg argued that this difference is actually due to the different effect that the haze of microscopic quantum activity has on each force. Their calculations showed that if this haze is penetrated by examining the forces not on everyday scales but as they act on distances of about a hundredth of a billionth of a billionth of a billionth(10-29) of a centimeter (a mere factor of ten thousand larger than the Planck length), the three nongravitational force strengths appear to become equal.

Although far removed from the realm of common experience, the high energy necessary to be sensitive to such small distances was characteristic of the roiling, hot early universe when it was about a thousandth of a trillionth of a trillionth of a trillionth (10-39) of a second old—when its temperature was on the order of 1028 Kelvin mentioned earlier. In somewhat the same way that a collection of disparate ingredients—pieces of metal, wood, rocks, minerals, and so on—all melt together and become a uniform, homogeneous plasma when heated to sufficiently high temperature, these theoretical works suggested that the strong, weak, and electromagnetic forces all merge into one grand force at such immense temperatures. This is shown schematically in Figure 7.1.6